

Synthetic Native Template

Redistributable test fixture — not
laboratory data

Representative content layout

- Native title/content placeholders

Progress / Previous Commitments

Commitment	NS101	planned	Owner: Gary	Due: 2026-09-10T09:00:00Z	H002
Commitment	NS201	planned	Owner: Gary	Due: 2026-09-24T00:00:00Z	

Dependencies	pressure fixture	Parallel:
True		
Dependencies	pressure fixture	Parallel:
True		

H001 | Hypothesis

Hypothesis | Bulk conductivity drives the positional signal gradient.

Falsifier | Matched conditions do not change the predicted CV or resistance.

Research question | 提升 bulk conductivity 是否足以降低位置依賴的訊號變異？

H001 | Problem

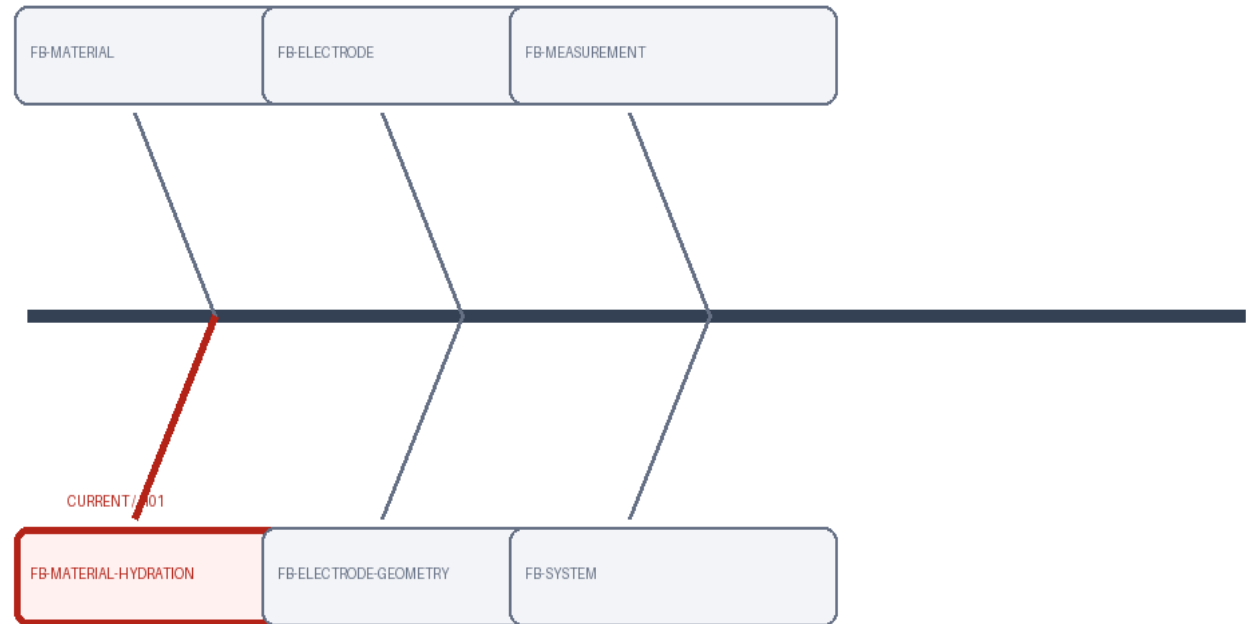
含水量與導電度存在位置梯度

導電度提高是否足以修復重複性
仍未知。

提升 bulk conductivity 是否
足以降低位置依賴的訊號變
異？

H001 | Total Fishbone / Research Map

FB-MATERIAL-HYDRATION



H001 | Observation → Literature → Mechanism

提升 bulk conductivity 是否足以降低位置依賴的訊號變異？

Observation | 位置依賴缺陷與訊號變異已觀察到。

Problem | 現有模型無法解釋此變異。

SYNTHETIC OBSERVATION

H001 | Literature → Mechanism → Strategy

Literature consensus | transport gradient 可產生位置效應。

Alternatives | interface contact remains alternative

Gap | 缺少控制比較隔離機制。

Implication | 需匹配條件後比較。

Mechanism | bulk transport

Strategy | 均化導電度

H001 | Experiment Design

EXP101 | IV: ['含水量']

Controls: ['原始配方']

N/replicates: {'replicates': 3, 'samples': 15}

Metrics/units: ['導電度 (mS/cm)']

Method: ['synthetic-four-probe']

Prediction: ['導電度提升']

Decision rule: {'go': 'mean conductivity increases $\geq 20\%$ ', 'partial_go': '10-20%', 'no_go': '<10%'}

H001 | Experiment Design

EXP102 | IV: ['位置']

Controls: ['原始位置分布']

N/replicates: {'replicates': 3, 'samples': 15}

Metrics/units: ['訊號 CV (%)']

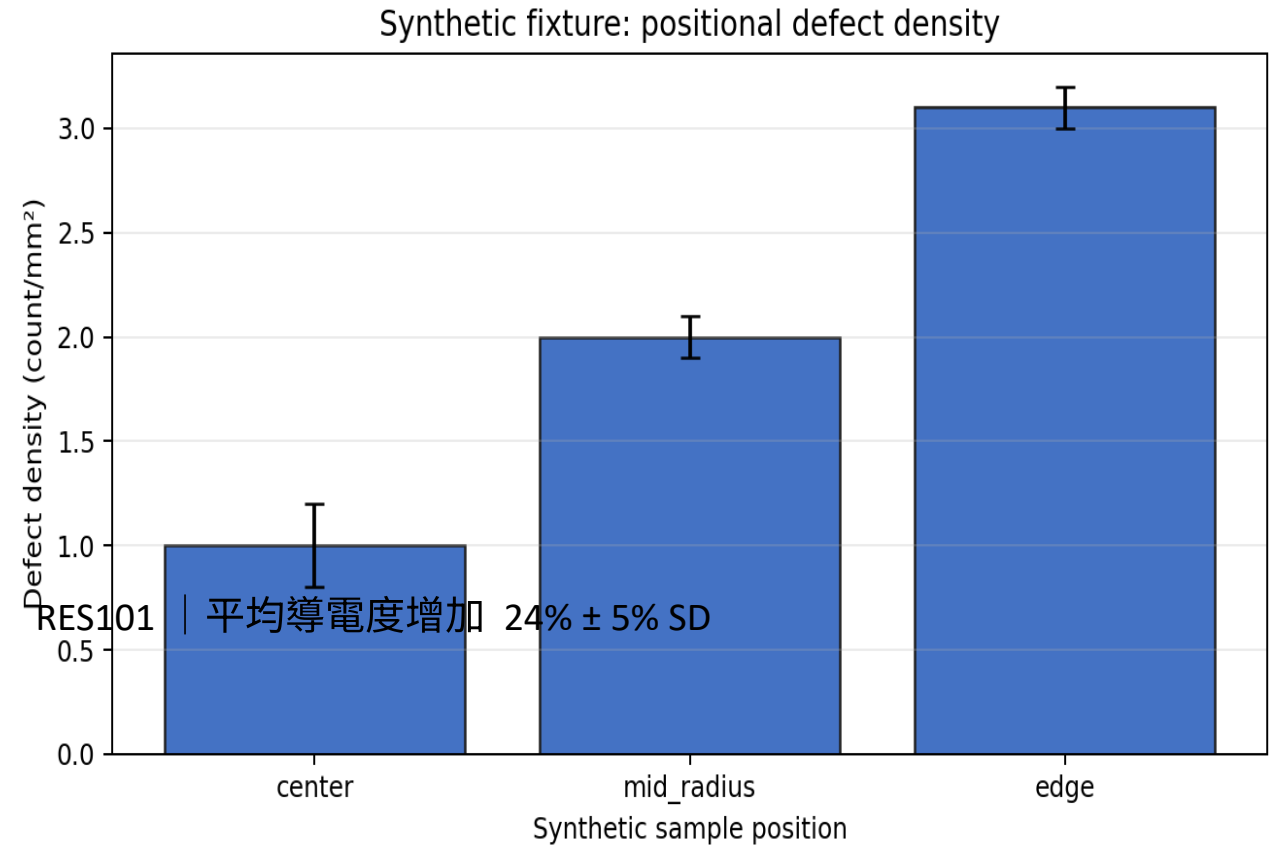
Method: ['synthetic-signal-test']

Prediction: ['CV 下降']

Decision rule: {'go': 'CV decreases $\geq 30\%$ ',
'partial_go': '10-30%', 'no_go': '<10%'}

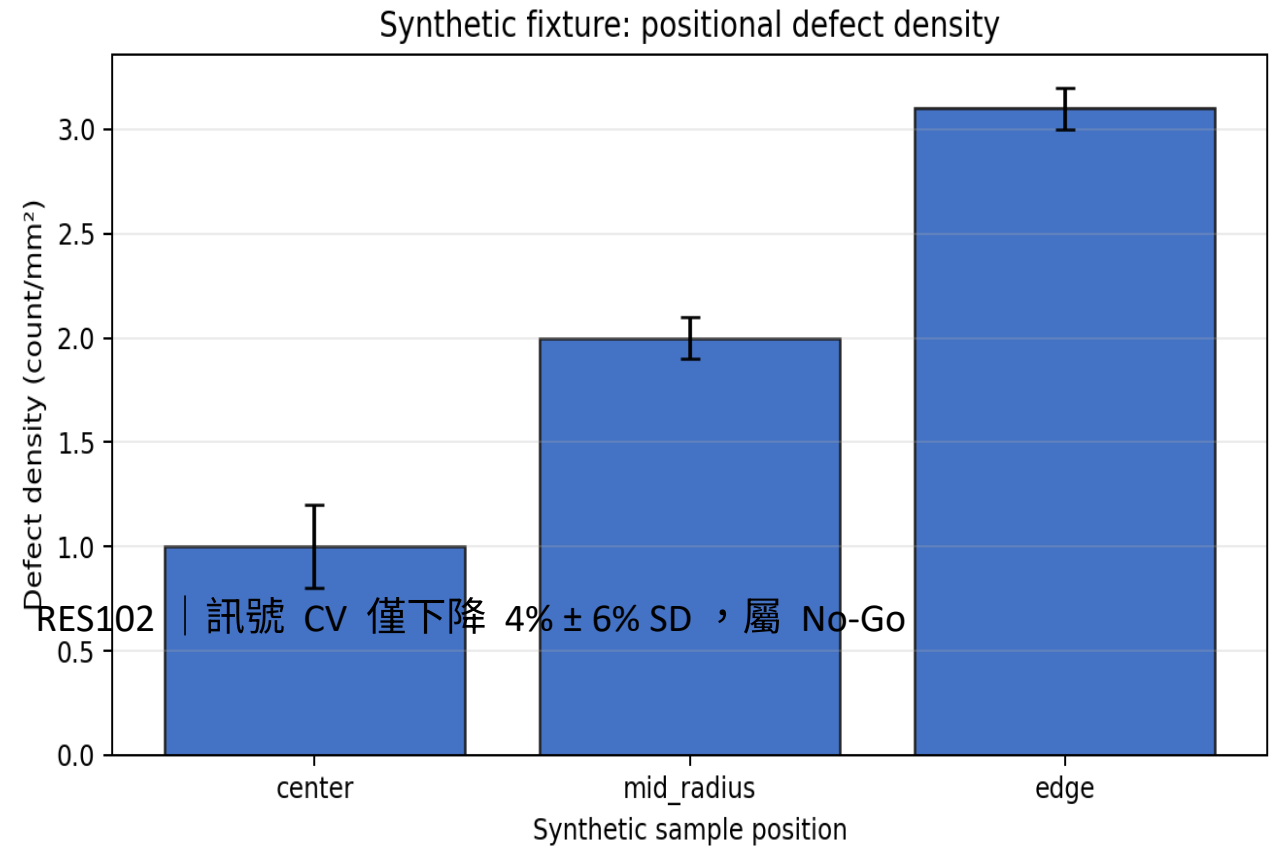
H001 | Result

RES101 | 平均導電度增加 $24\% \pm 5\%$ SD



H001 | Result

RES102 | 訊號 CV 僅下降 4% ± 6% SD , 屬 No-Go



H001 | Integrated Discussion

RES101

RES102

Mechanism assessment | bulk conductivity 是部分機制，不足以解釋 interface-sensitive variation 。

Alternatives | 接觸電阻；局部壓力差；電極幾何

Remaining uncertainty | contact interface 是否為主導因素仍需匹配導電度後檢驗。

H001 | Layer Summary / Decision

Answered | bulk conductivity 影響缺陷梯度，但不足以單獨解釋重複性。

Hypothesis status |
partial_support

Decision | Partial-Go: Bulk conductivity is contributory but insufficient.

interface contact 的貢獻尚未量化。

Next question | 匹配 bulk conductivity 後，contact resistance 是否主導變異？
Next Step | NS101

H001 | Hypothesis Transition

Previous hypothesis | C101

Key results | RES101, RES102

New observation | E104

New hypothesis | C201

Not explained | 導電度增加後訊號重複性沒有改善。

Therefore | 剩餘變異隨電極接觸位置改變，因此核心解釋由 bulk transport 轉向 interface contact 。

H002 | Hypothesis

Hypothesis | Contact resistance drives instability at low pressure.

Falsifier | Matched conditions do not change the predicted CV or resistance.

Research question | 低接觸壓力下，contact resistance 是否主導訊號不穩定？

H002 | Problem

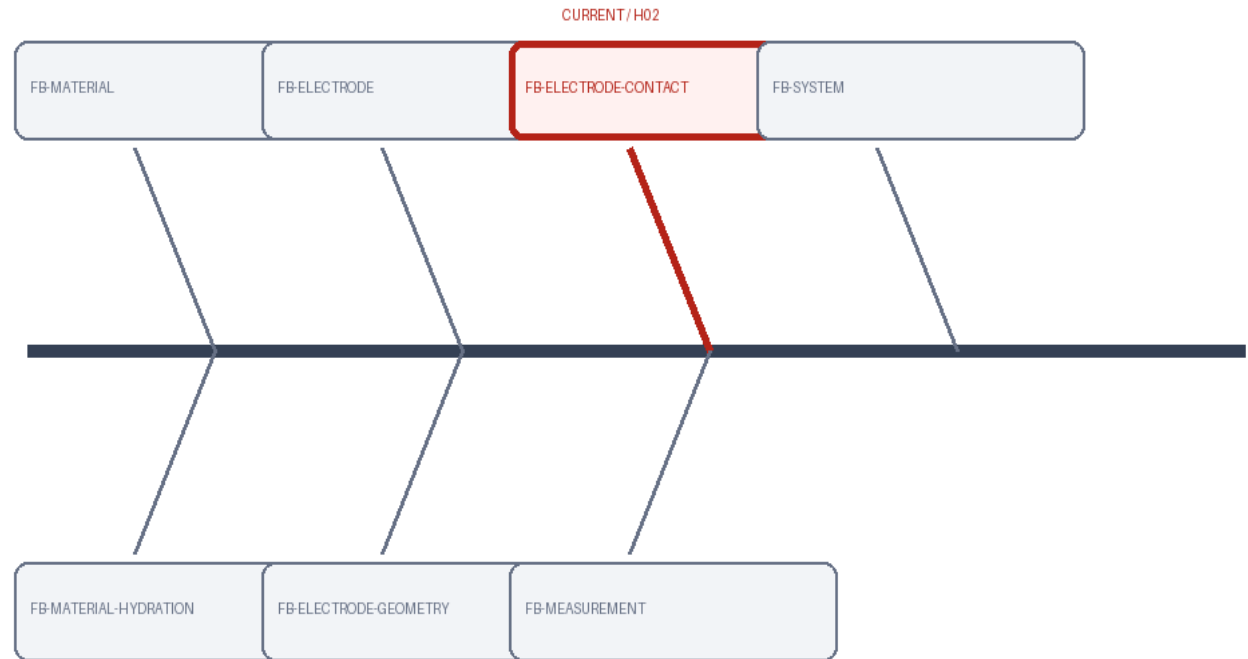
H01 支持導電度梯度存在 ; H01
未支持導電度單獨主導重複性

bulk transport 與 interface
contact 的相對貢獻尚未區分。

低接觸壓力下， contact
resistance 是否主導訊號不穩定？

H002 | Total Fishbone / Research Map

FB-ELECTRODE-CONTACT



H002 | Observation → Literature → Mechanism

Observation 位置依賴缺陷與訊號變異已觀察到。	Literature consensus transport gradient可產生位置效應。	Mechanism contact resistance	匹配導電度並改變接觸壓力
Problem 現有模型無法解釋此變異。	Alternatives interface contact remains alternative		
	Gap 缺少控制比較隔離機制。		
	Implication 需匹配條件後比較。		

低接觸壓力下， contact resistance 是否主導訊號不穩定？

Registered observation visual.

H002 | Results

IV: contact pressure

Controls: bulk conductivity; 電極幾何

Baselines: low-pressure control

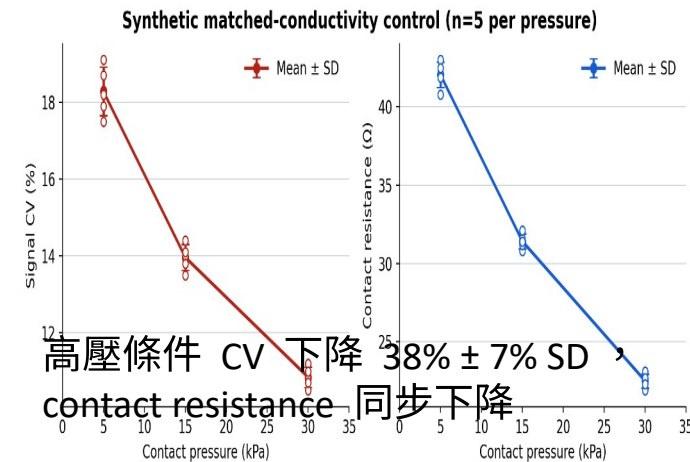
N/replicates: replicates: 5; samples: 15

Metrics/units: 訊號 CV (%); contact resistance (ohm)

Method: synthetic-pressure-fixture

Prediction: CV and resistance decrease

go: both metrics
improve;
partial_go: one
metric improves;
no_go: neither
improves



高壓條件 CV 下降 38% ± 7% SD , contact resistance 同步下降

H002 | Layer Summary / Decision

RES201

None identified

Cross-experiment pattern | 匹配導電度後，提高接觸壓力明顯降低變異。

Mechanism assessment | contact resistance 在低壓條件下主導不穩定。

Alternatives | 電極幾何仍可能調節壓力分布

Discussion uncertainty | 高壓長期循環是否造成新的 wear mechanism 尚未知。

Summary unresolved | 高壓循環的耐久性與 wear 尚未測試。

Hypothesis status | supported

Decision | Go: Contact-pressure control discriminates the interface mechanism.

Next question | 接觸壓力改善能否在長期循環下維持？

Next Step | NS201